Missionaries and Cannibals

Problem model, counting conventions and report scope

STATE

(M,C,B): missionaries and cannibals on the left bank; B is the boat side, L or R. Right-bank counts are 3-M and 3-C.

INITIAL STATE

(3,3,L): all three missionaries, all three cannibals and the boat start on the left.

ACTIONS

Boat loads, in fixed order: 1M, 2M, 1C, 2C, 1M+1C. Every crossing changes the boat side.

SEARCH RESULT

Visited / dequeued states: 15
Unique discovered, including the initial state: 15
Attempted transitions: 70
Rejected invalid / repeated: 40 / 16
Shortest solution depth: 11

CONSTRAINTS

The boat carries 1-2 available passengers. On either bank with missionaries, cannibals may not outnumber them. A bank with zero missionaries may contain cannibals.

GOAL

(0,0,R): everyone and the boat are on the right bank.

STRATEGY

Unit-cost BFS with discovered states. Invalid loads and unsafe states are rejected; duplicates are not enqueued.

PAGES THAT FOLLOW

The next 15 pages show every BFS expansion or goal check up to the first dequeued goal, followed by the final route. Every page retains the tree generated so far, starting at the root, and enlarges the current expansion below. Repeated states are yellow attempts and never enter the queue twice. The overview uses fixed positions; zoom the vector PDF to inspect small nodes.

COUNTING NOTE

Statistics apply only to this instance, action order and goal-test timing. Attempted-transition nodes are not the same as unique discovered states. Green nodes identify the returned route retrospectively.

BFS step 1 of 15: expand node #0

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

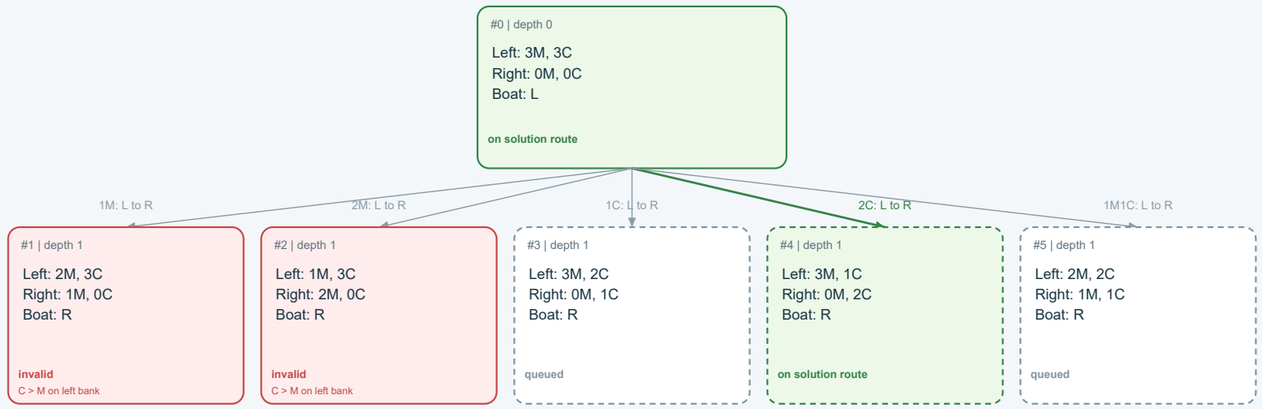
Processed: 1 | Unique discovered: 4 | Queue after step: 3 | Attempts so far: 5

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 3 new state(s), 0 repeated state(s), 2 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

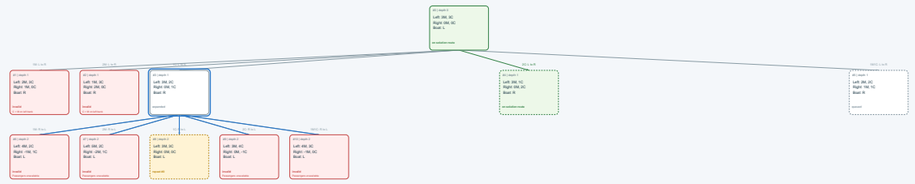
BFS step 2 of 15: expand node #3

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

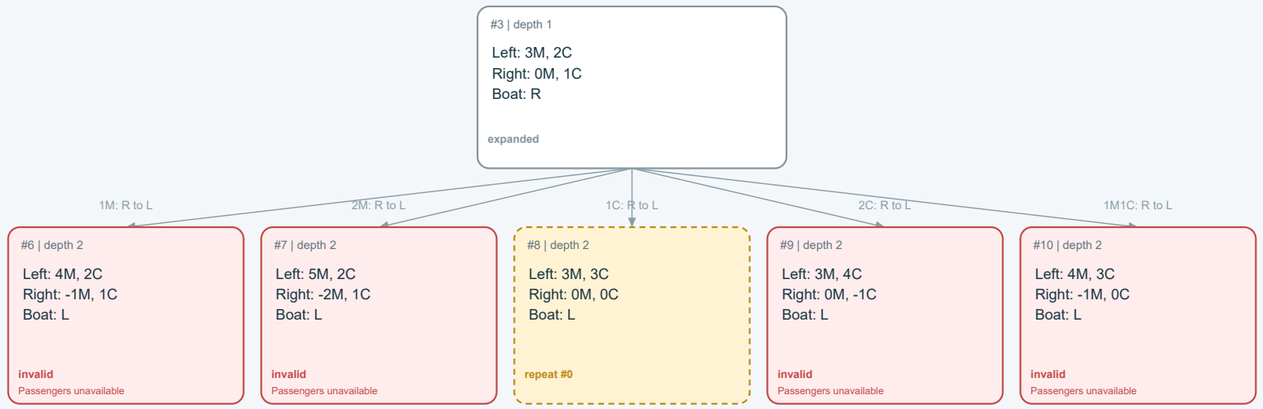
Processed: 2 | Unique discovered: 4 | Queue after step: 2 | Attempts so far: 10

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 0 new state(s), 1 repeated state(s), 4 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

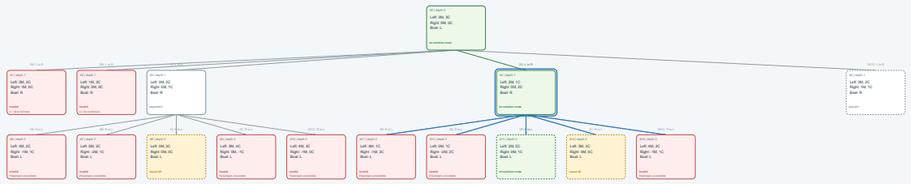
BFS step 3 of 15: expand node #4

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

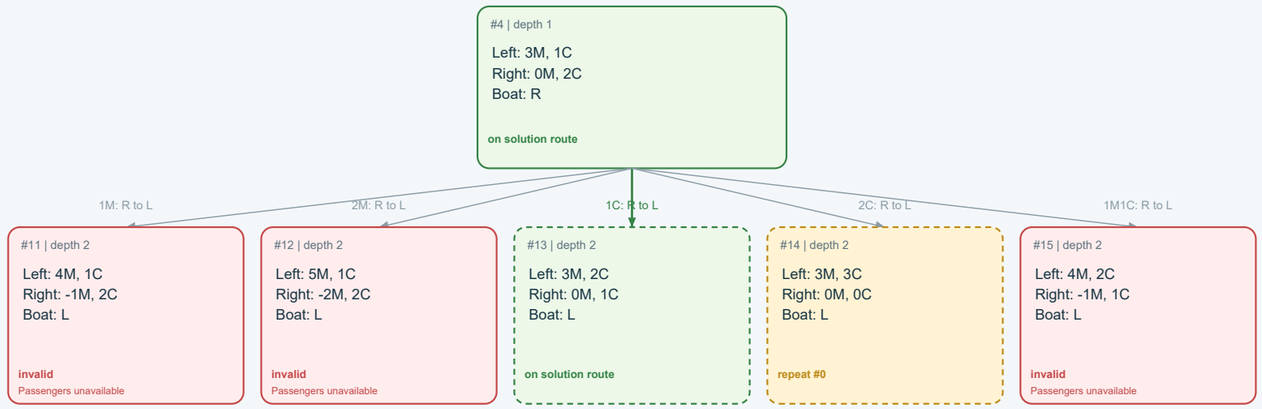
Processed: 3 | Unique discovered: 5 | Queue after step: 2 | Attempts so far: 15

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

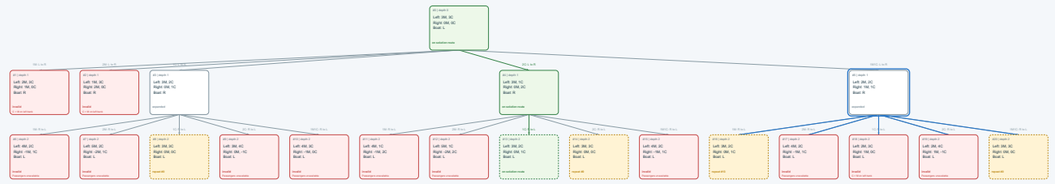
BFS step 4 of 15: expand node #5

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

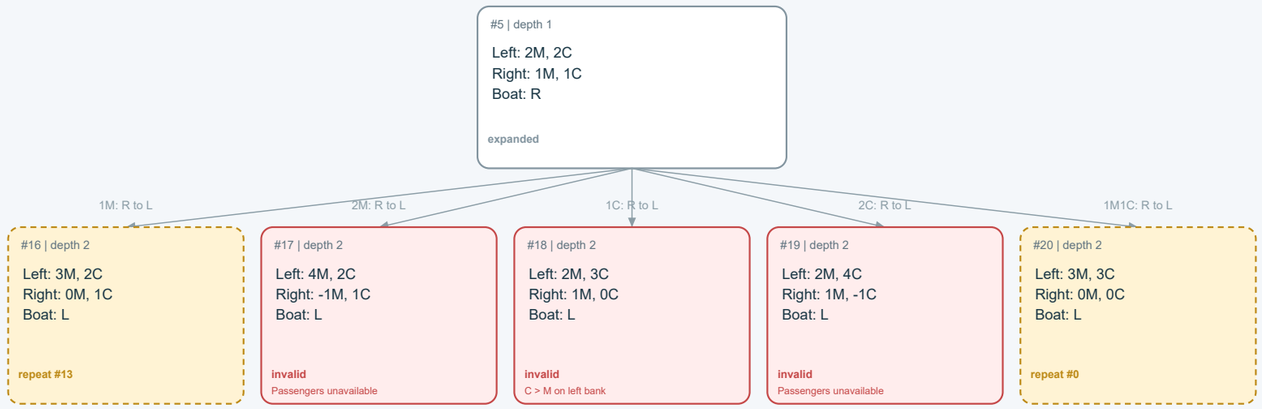
Processed: 4 | Unique discovered: 5 | Queue after step: 1 | Attempts so far: 20

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 0 new state(s), 2 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

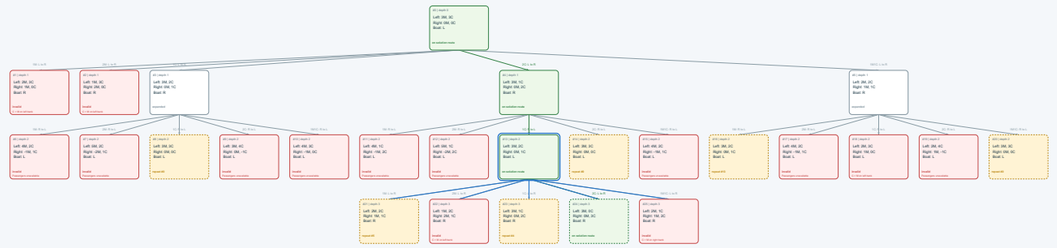
BFS step 5 of 15: expand node #13

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

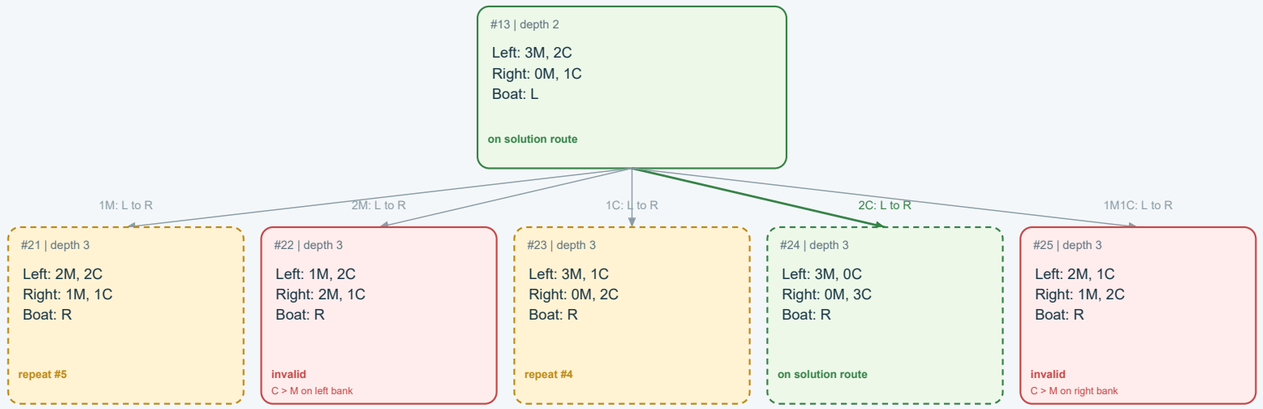
Processed: 5 | Unique discovered: 6 | Queue after step: 1 | Attempts so far: 25

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 2 repeated state(s), 2 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

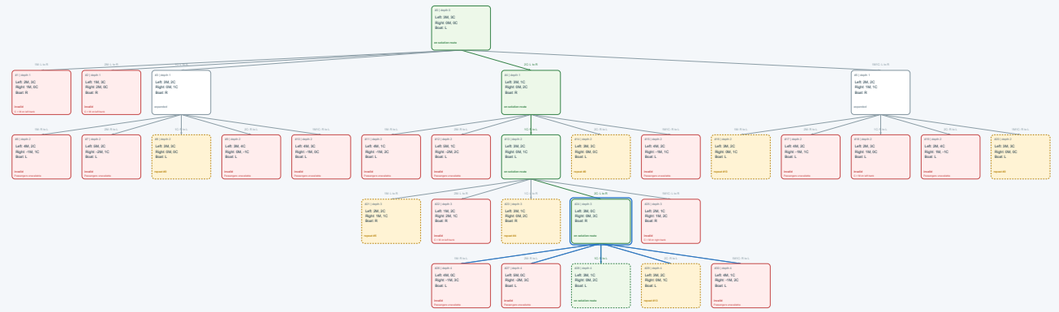
BFS step 6 of 15: expand node #24

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

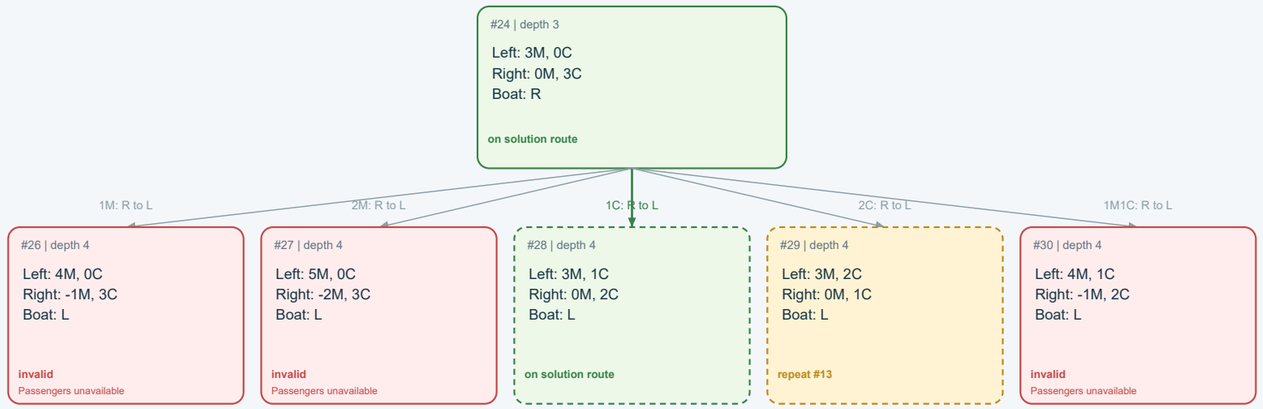
Processed: 6 | Unique discovered: 7 | Queue after step: 1 | Attempts so far: 30

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

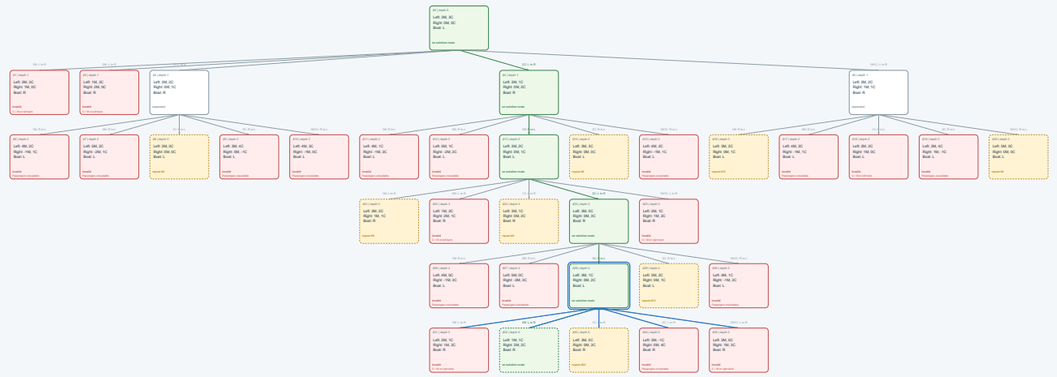
BFS step 7 of 15: expand node #28

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

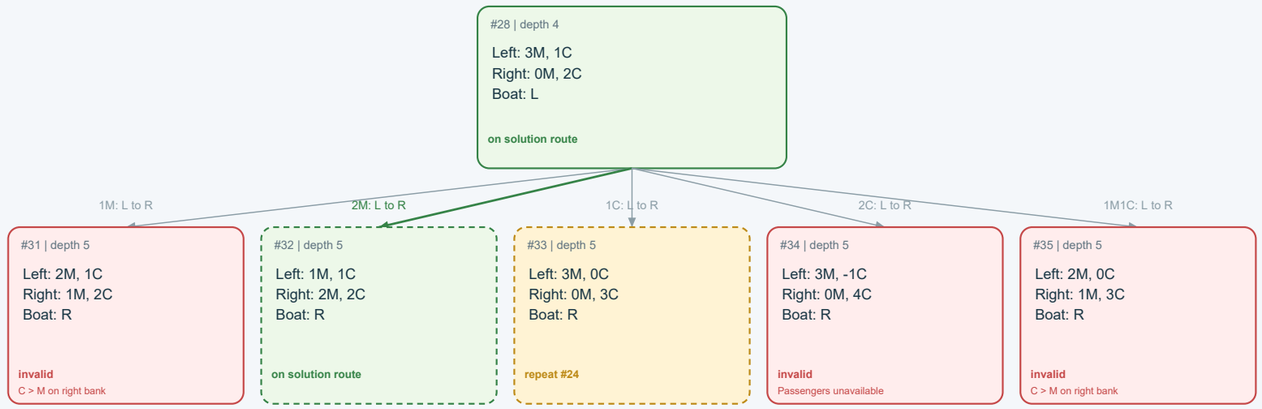
Processed: 7 | Unique discovered: 8 | Queue after step: 1 | Attempts so far: 35

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

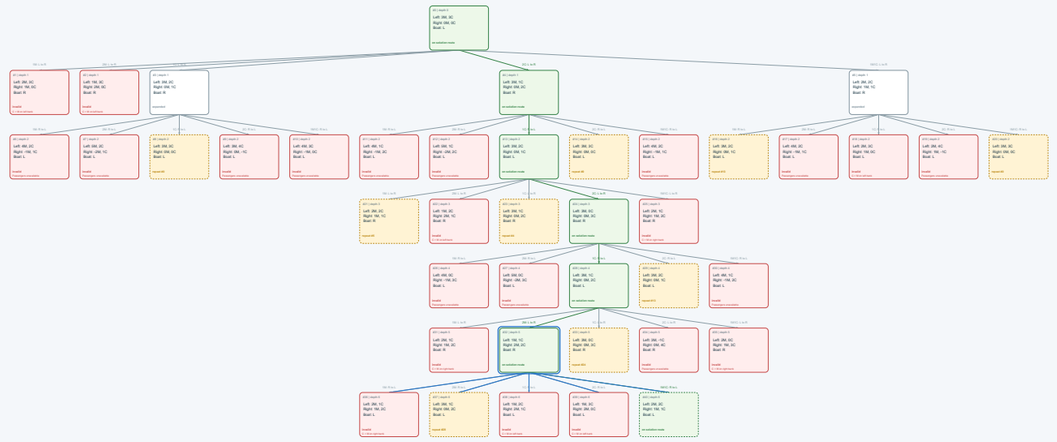
BFS step 8 of 15: expand node #32

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

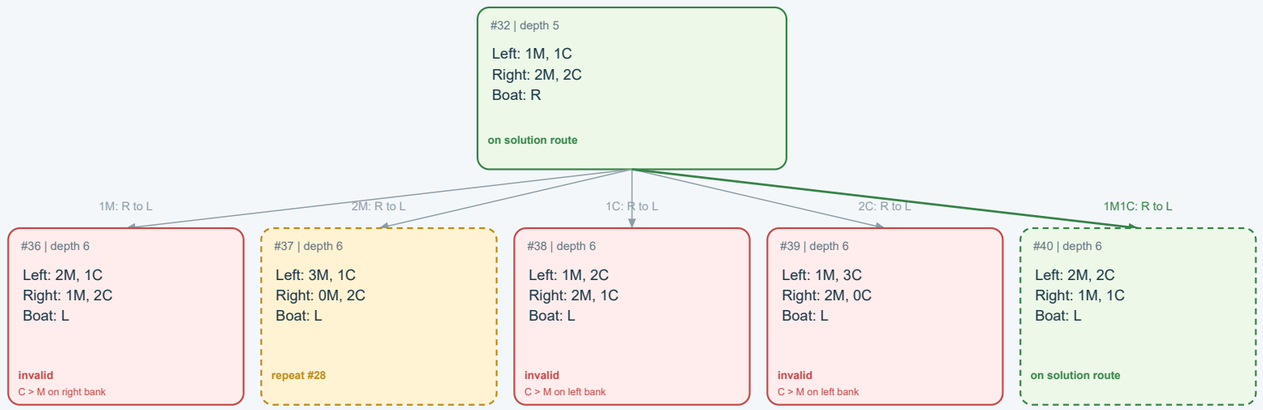
Processed: 8 | Unique discovered: 9 | Queue after step: 1 | Attempts so far: 40

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

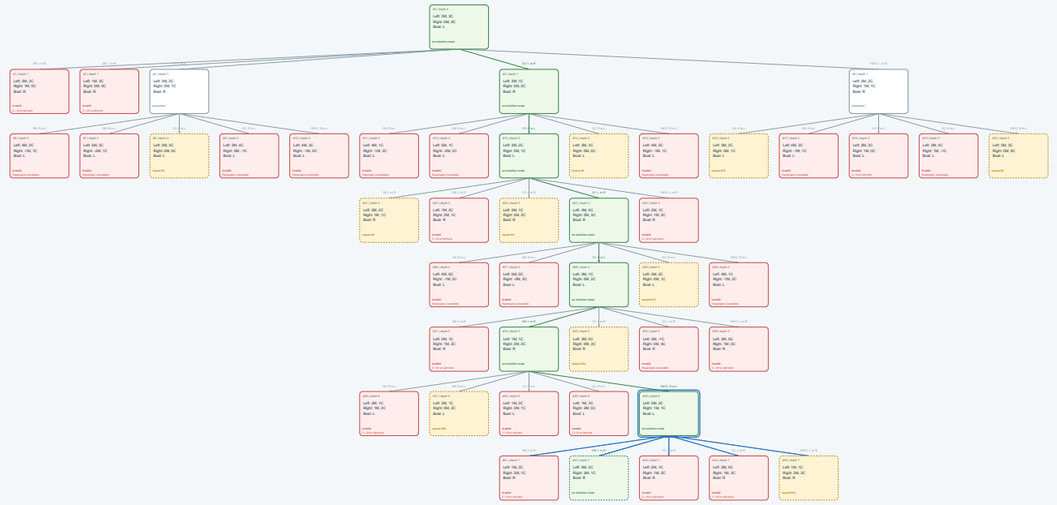
BFS step 9 of 15: expand node #40

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

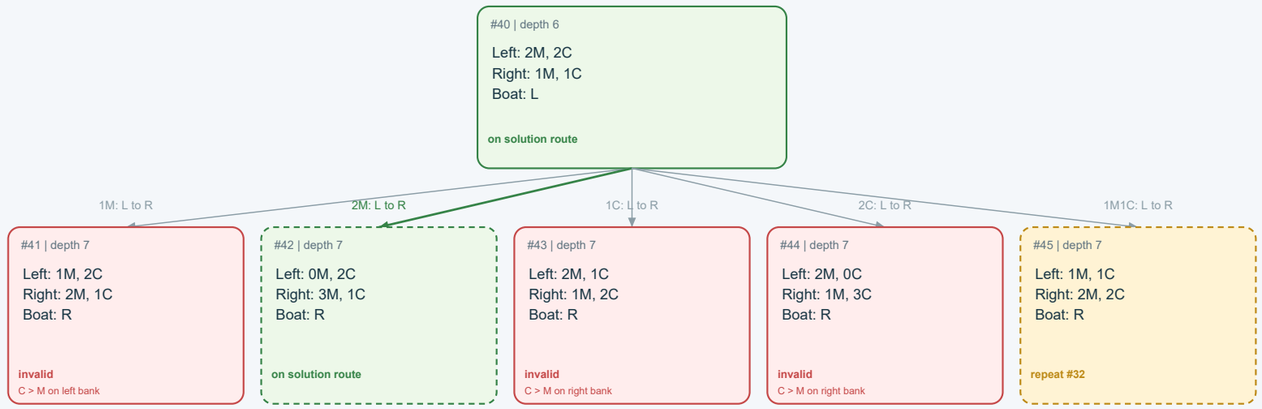
Processed: 9 | Unique discovered: 10 | Queue after step: 1 | Attempts so far: 45

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

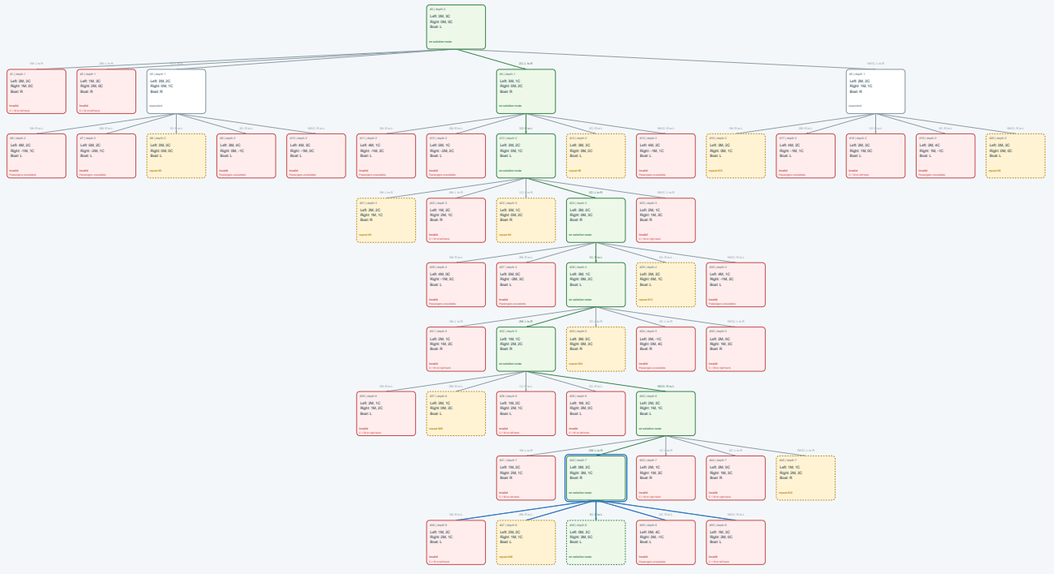
BFS step 10 of 15: expand node #42

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

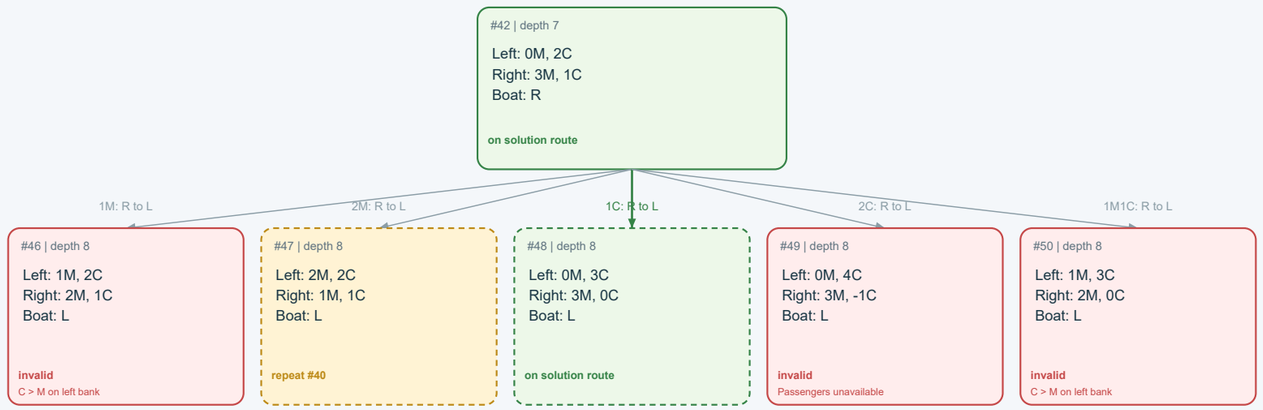
Processed: 10 | Unique discovered: 11 | Queue after step: 1 | Attempts so far: 50

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

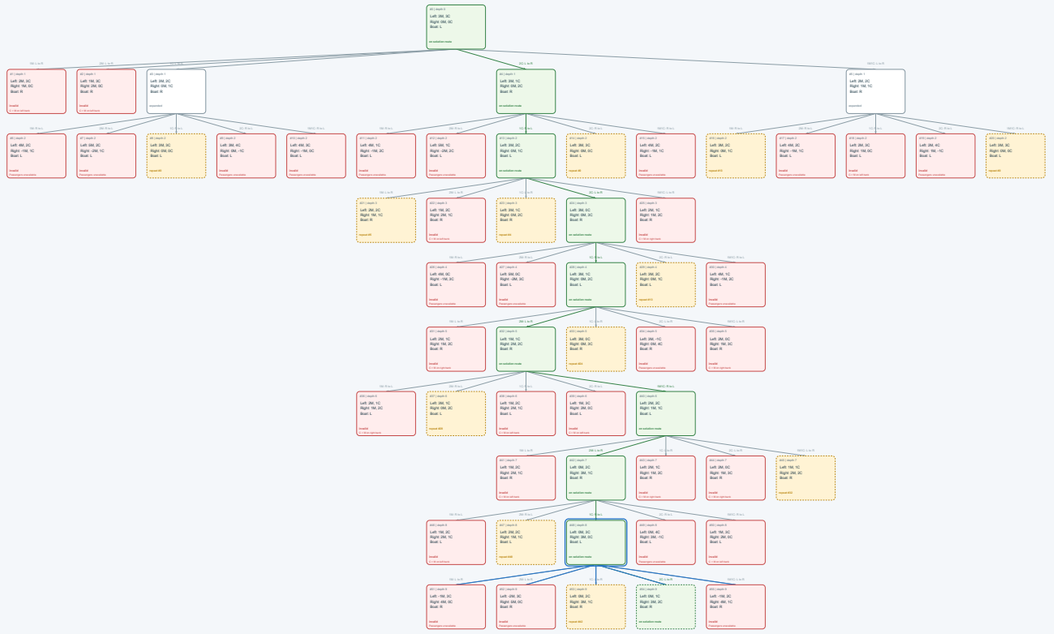
BFS step 11 of 15: expand node #48

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

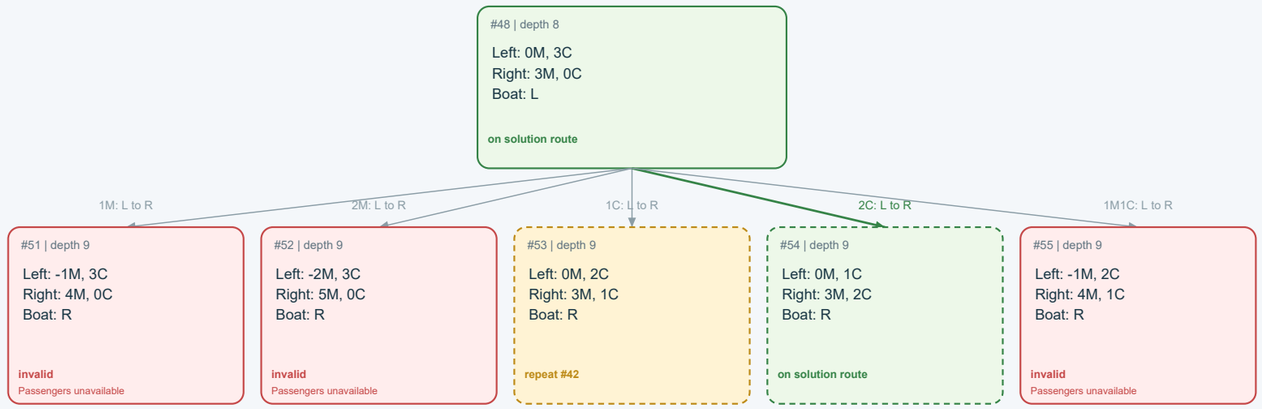
Processed: 11 | Unique discovered: 12 | Queue after step: 1 | Attempts so far: 55

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

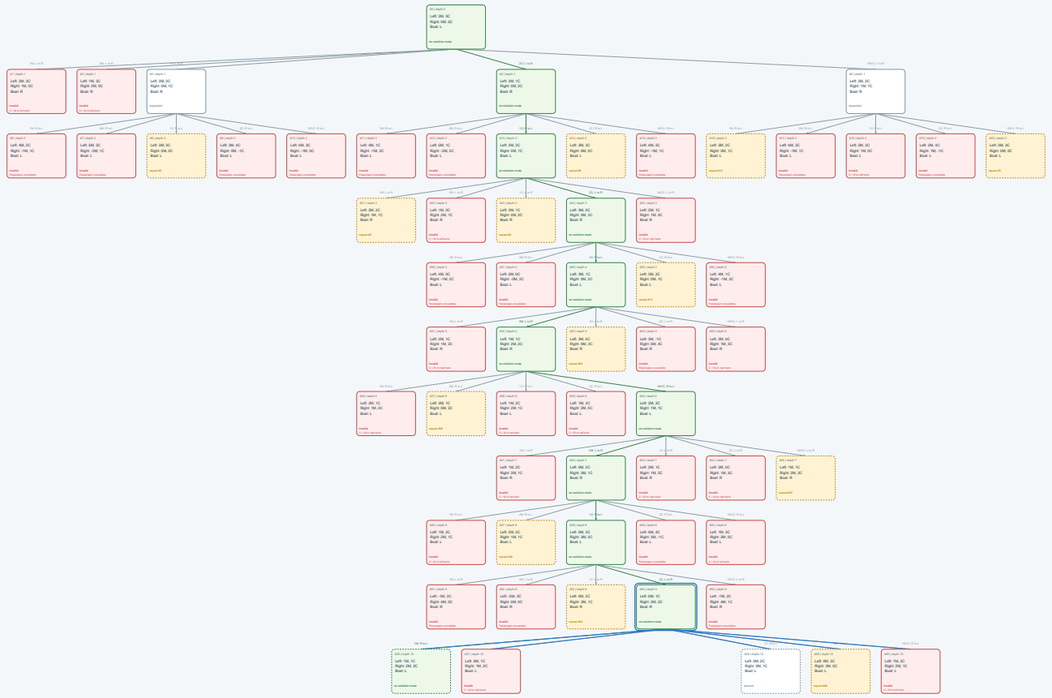
BFS step 12 of 15: expand node #54

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

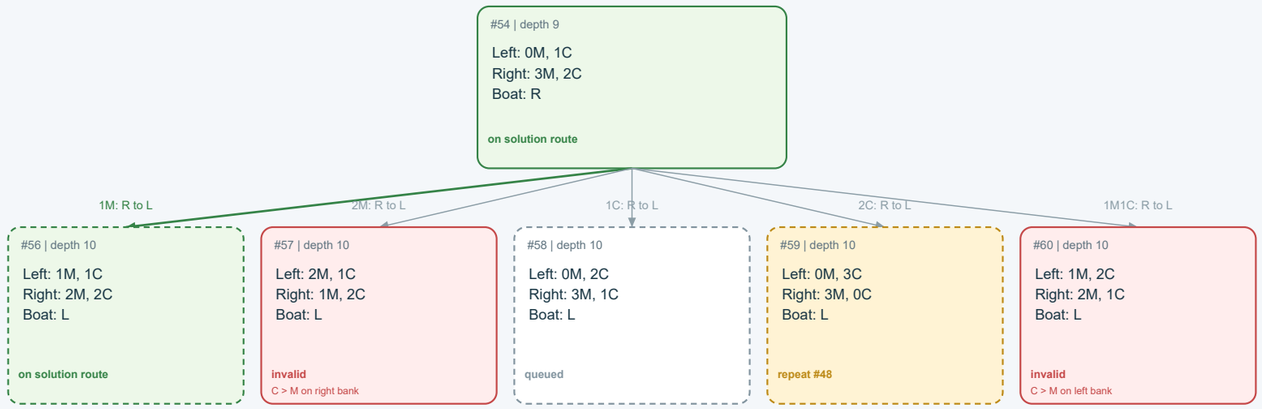
Processed: 12 | Unique discovered: 14 | Queue after step: 2 | Attempts so far: 60

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 2 new state(s), 1 repeated state(s), 2 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

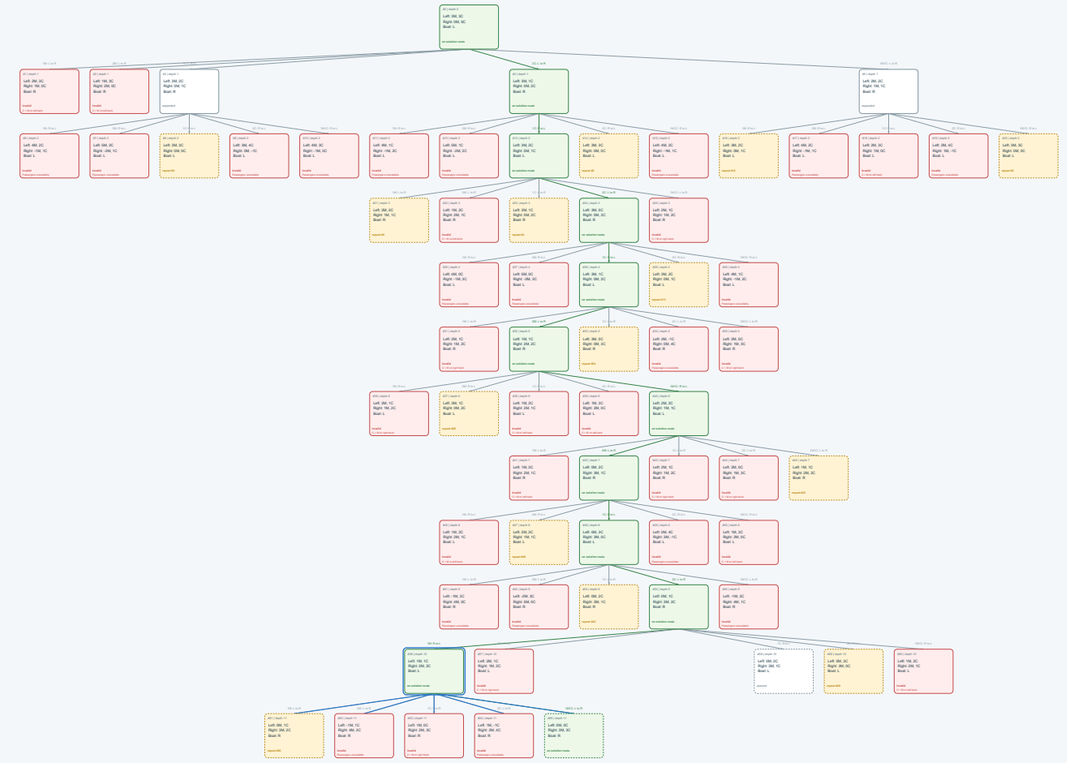
BFS step 13 of 15: expand node #56

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

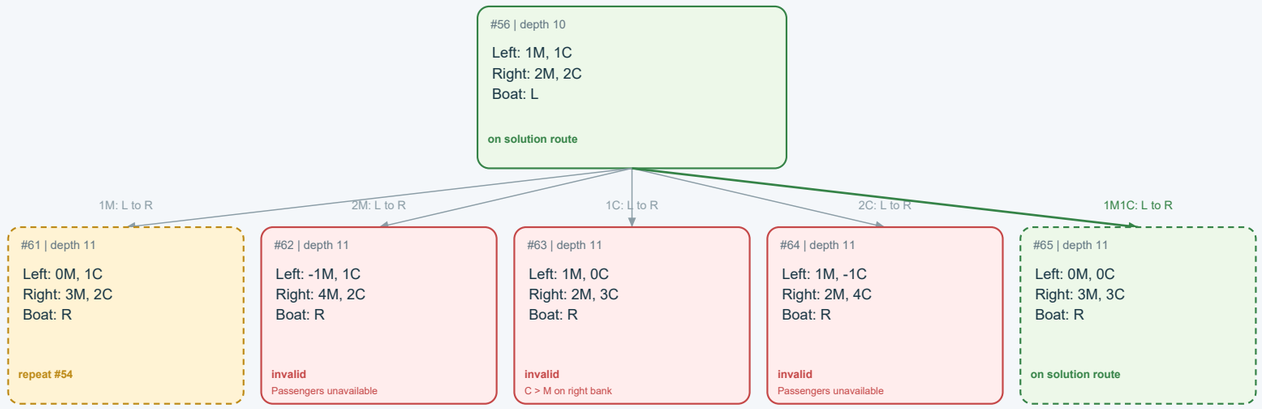
Processed: 13 | Unique discovered: 15 | Queue after step: 2 | Attempts so far: 65

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 1 new state(s), 1 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

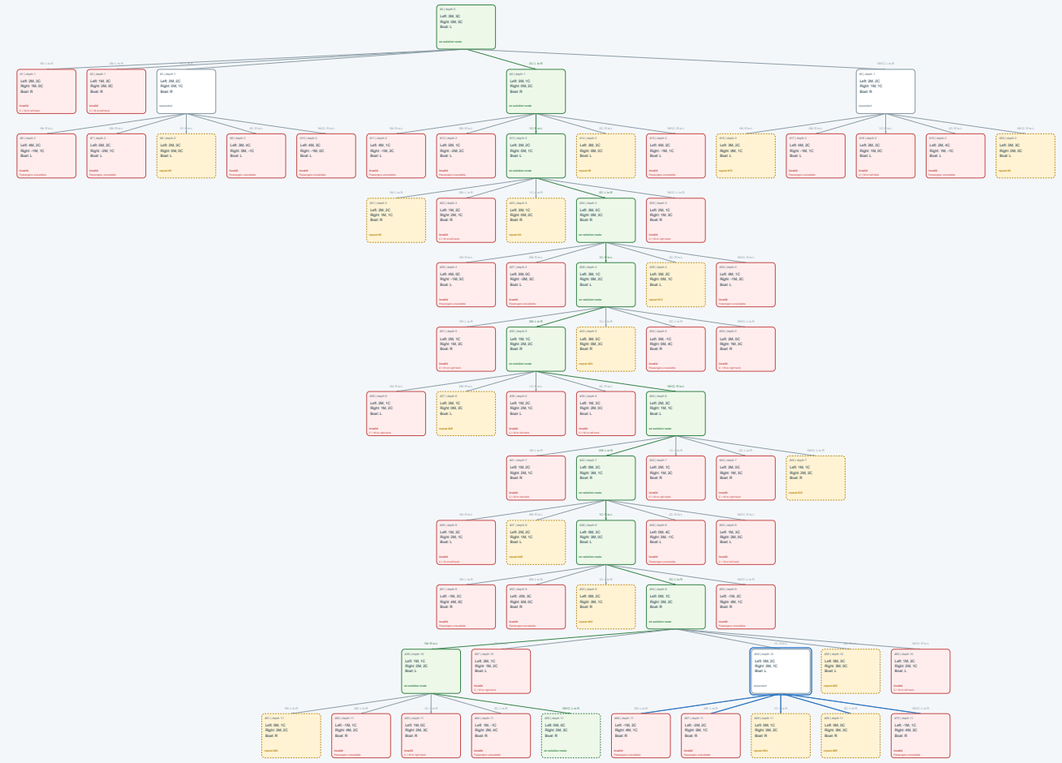
BFS step 14 of 15: expand node #58

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

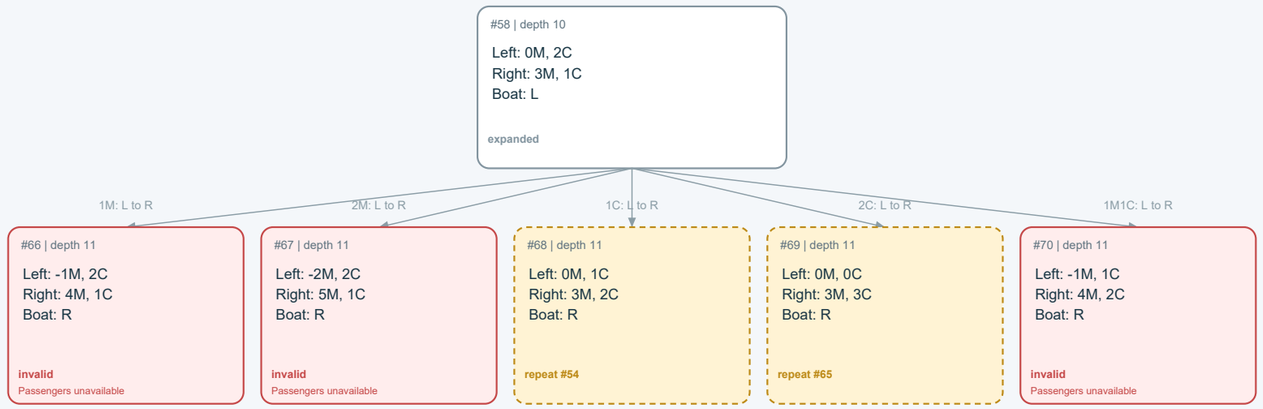
Processed: 14 | Unique discovered: 15 | Queue after step: 1 | Attempts so far: 70

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 5 moves: 0 new state(s), 2 repeated state(s), 3 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

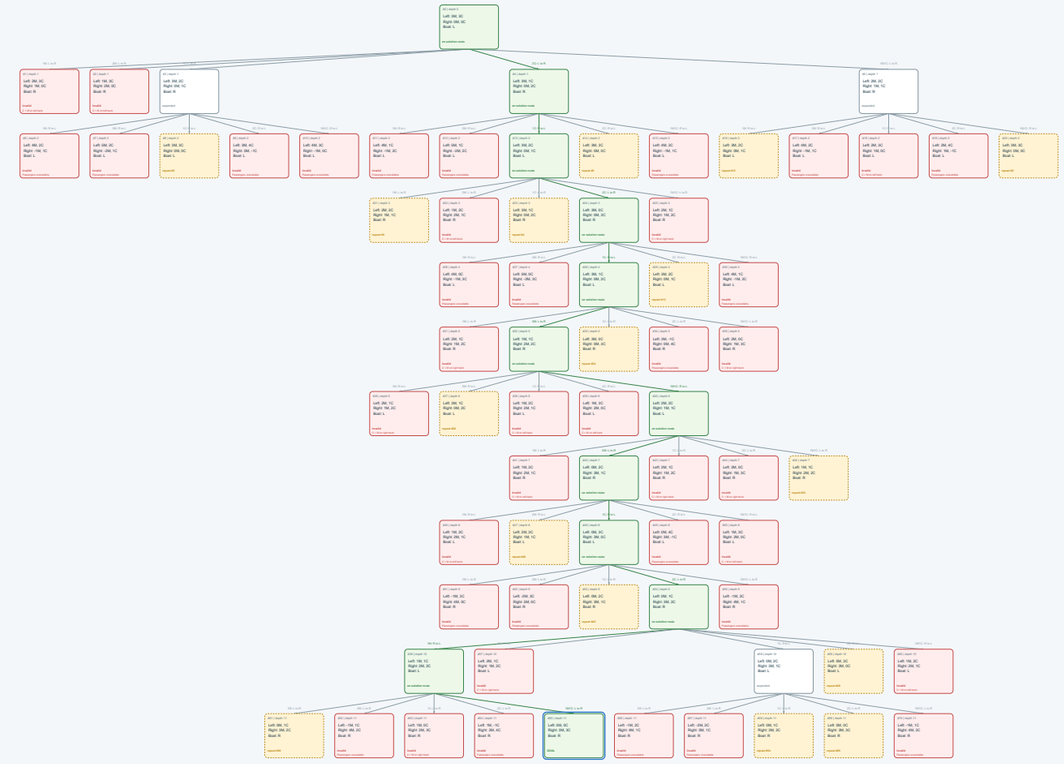
BFS step 15 of 15: goal check

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

Processed: 15 | Unique discovered: 15 | Queue after step: 0 | Attempts so far: 70

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#65 | depth 11

Left: 0M, 0C
Right: 3M, 3C
Boat: R

GOAL

The goal predicate is true. Search stops here; any queued alternatives are not needed to establish this solution.

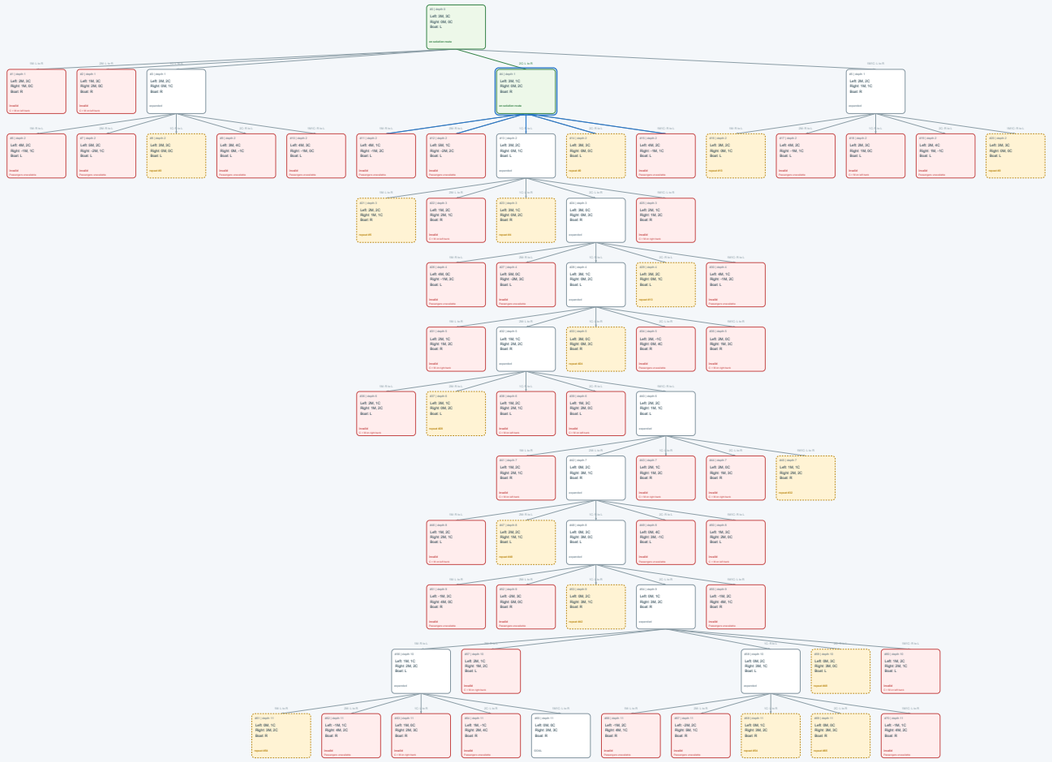
Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Final solution - move 1 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



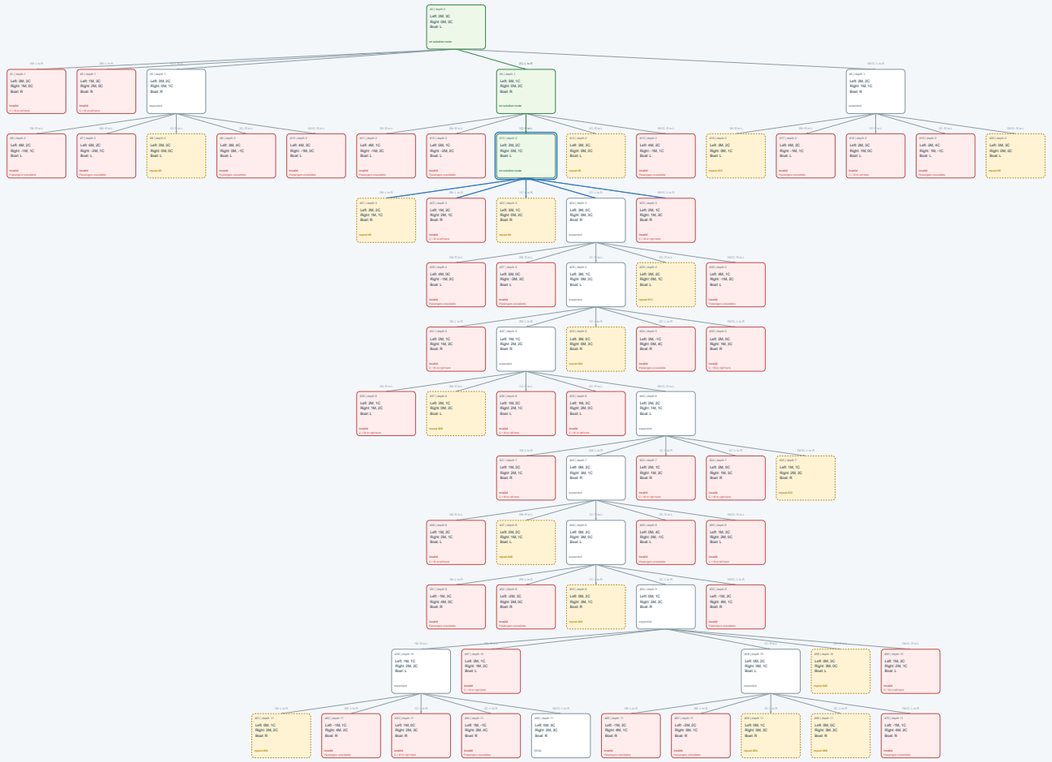
Move 2 cannibals from left to right. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 2 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



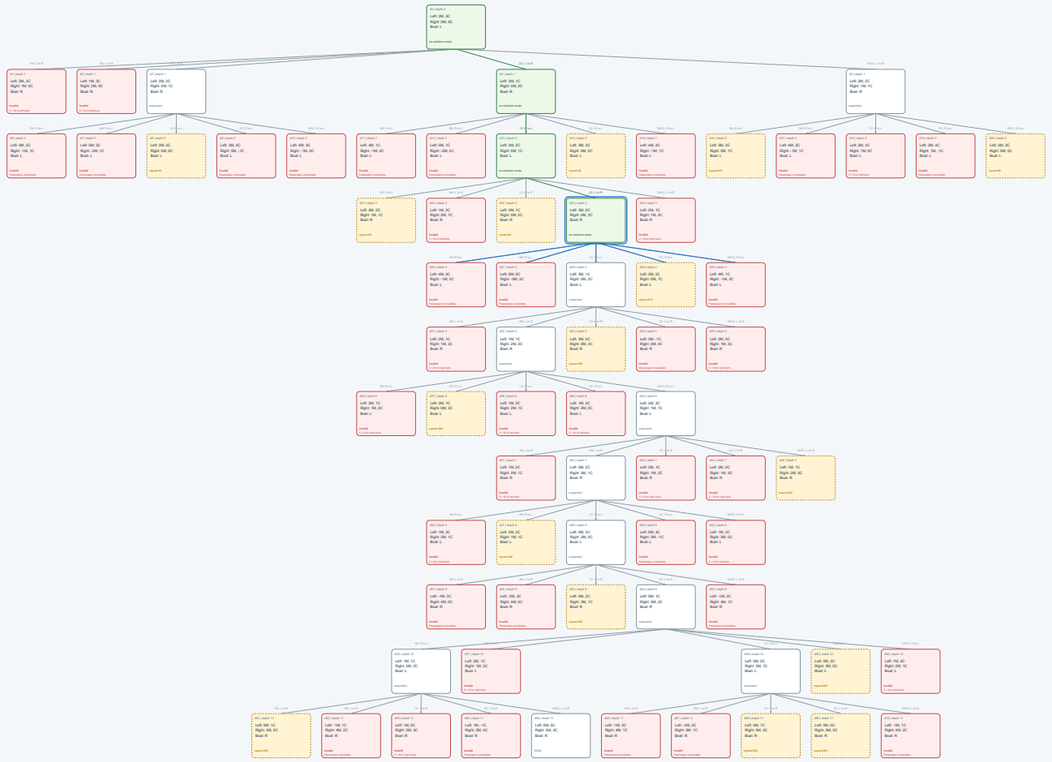
Move 1 cannibal from right to left. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 3 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#13 | depth 2

Left: 3M, 2C
Right: 0M, 1C
Boat: L

on solution route

2C: L to R

#24 | depth 3

Left: 3M, 0C
Right: 0M, 3C
Boat: R

on solution route

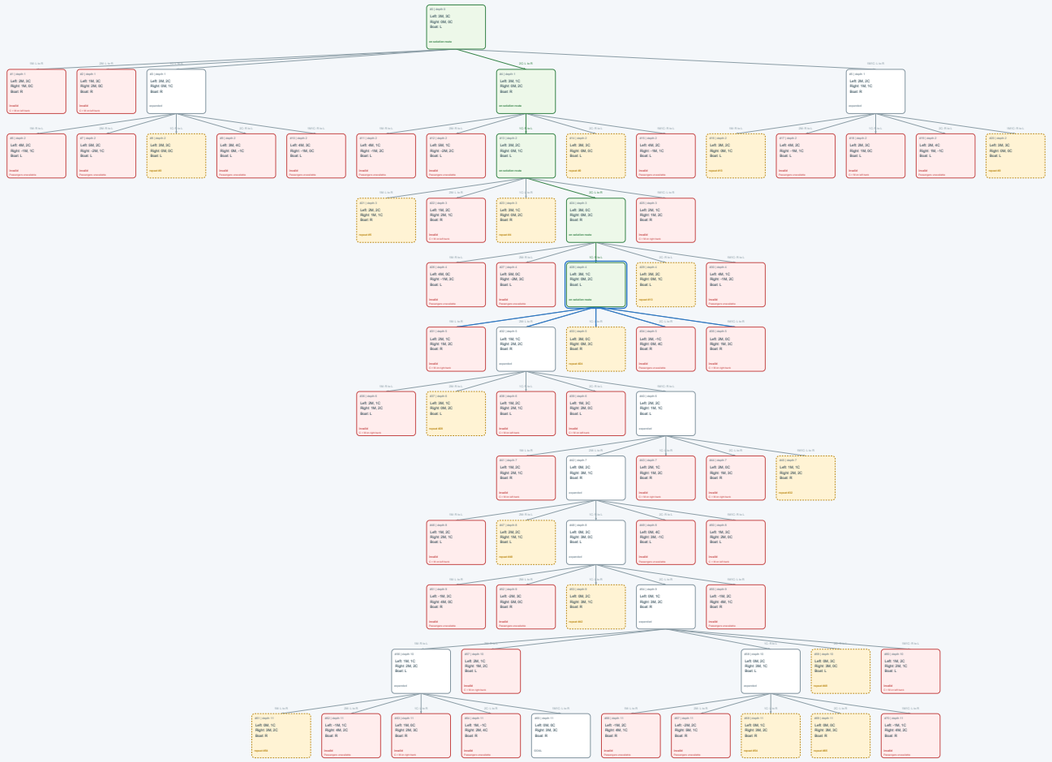
Move 2 cannibals from left to right. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 4 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#24 | depth 3

Left: 3M, 0C
Right: 0M, 3C
Boat: R

on solution route

1C: R to L

#28 | depth 4

Left: 3M, 1C
Right: 0M, 2C
Boat: L

on solution route

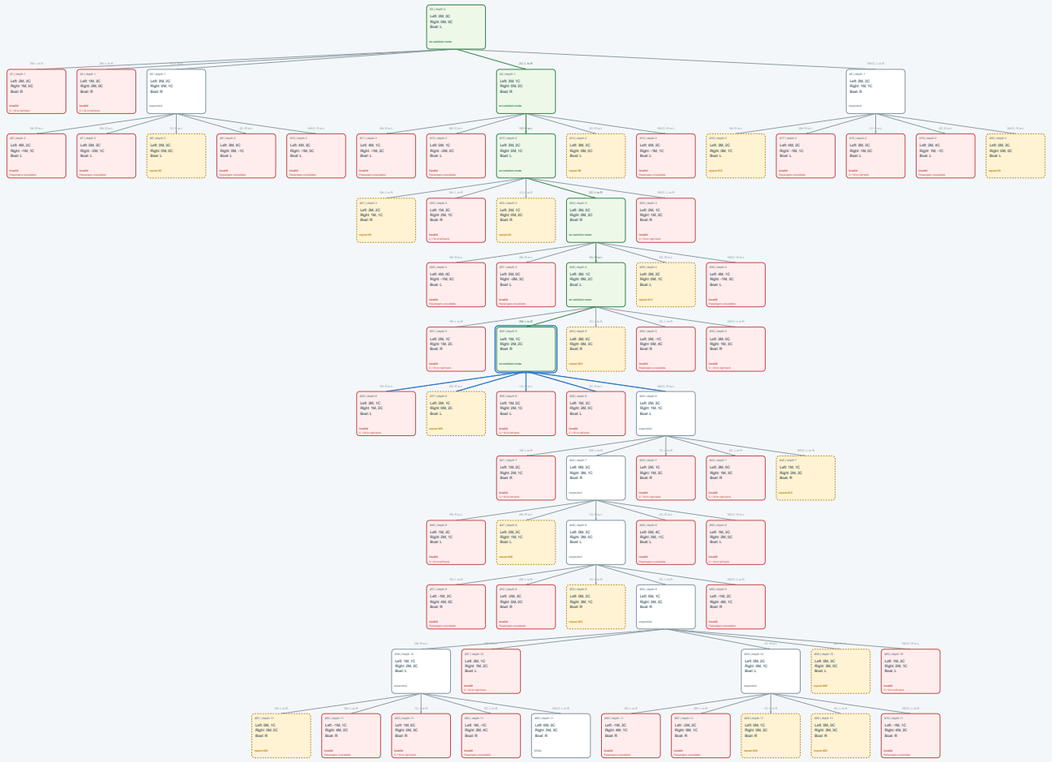
Move 1 cannibal from right to left. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 5 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#28 | depth 4

Left: 3M, 1C
Right: 0M, 2C
Boat: L

on solution route

2M: L to R

#32 | depth 5

Left: 1M, 1C
Right: 2M, 2C
Boat: R

on solution route

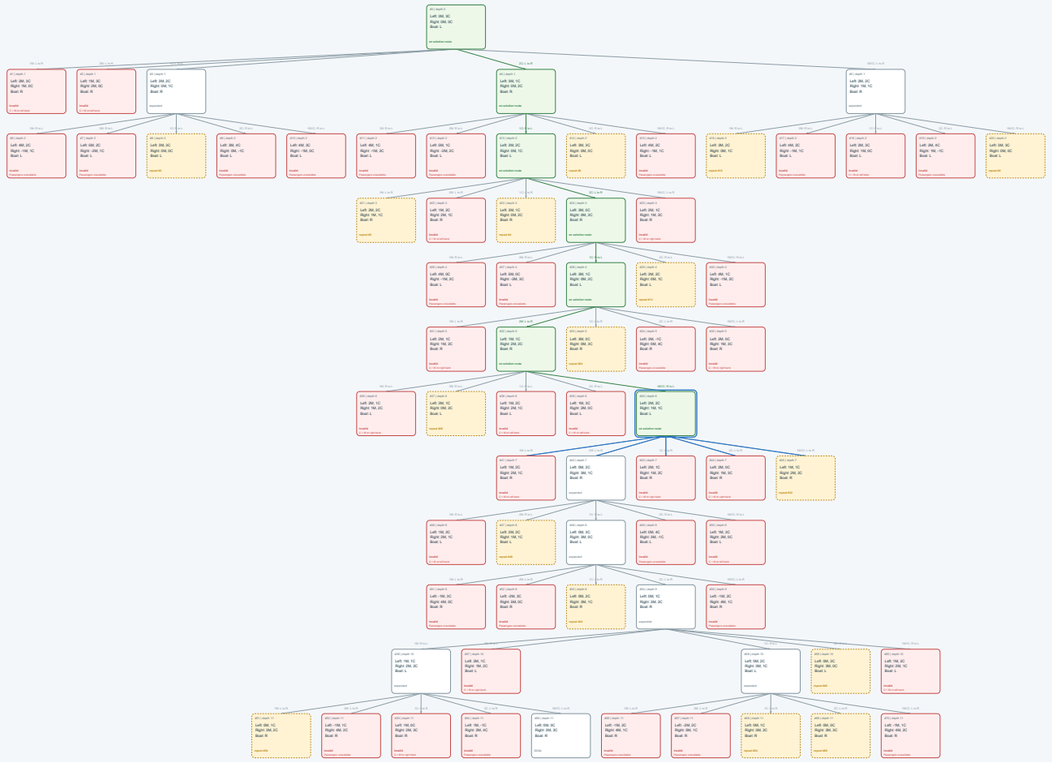
Move 2 missionaries from left to right. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 6 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



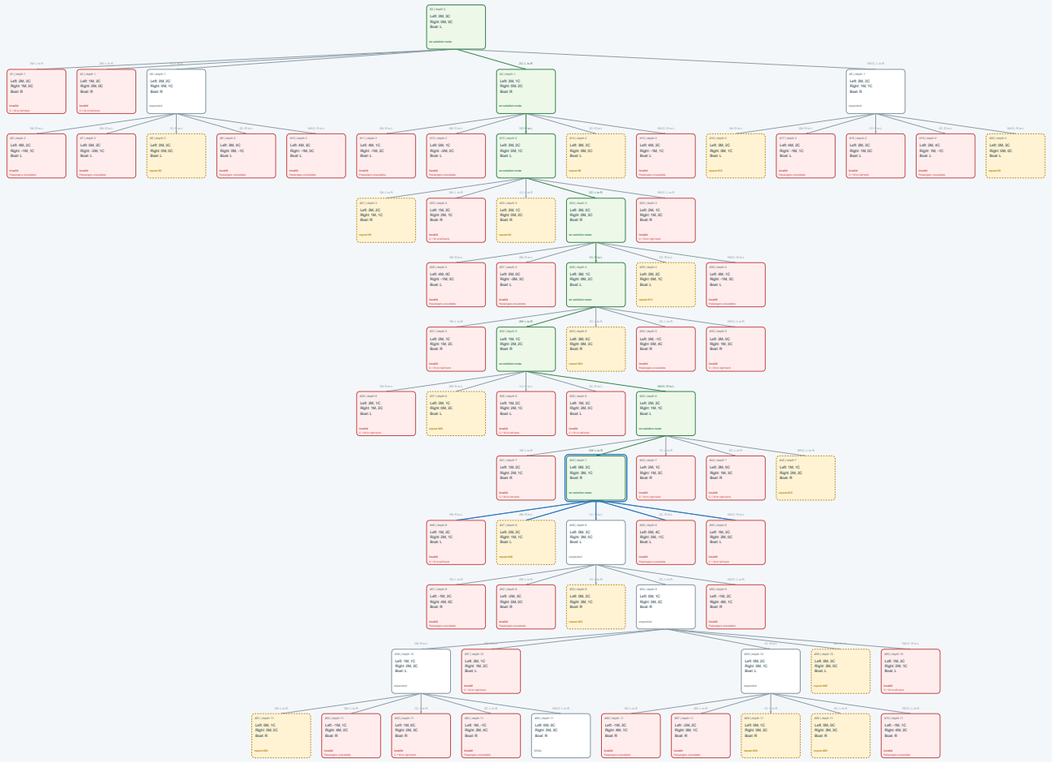
Move 1 missionary and 1 cannibal from right to left. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 7 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#40 | depth 6

Left: 2M, 2C
Right: 1M, 1C
Boat: L

on solution route

2M: L to R

#42 | depth 7

Left: 0M, 2C
Right: 3M, 1C
Boat: R

on solution route

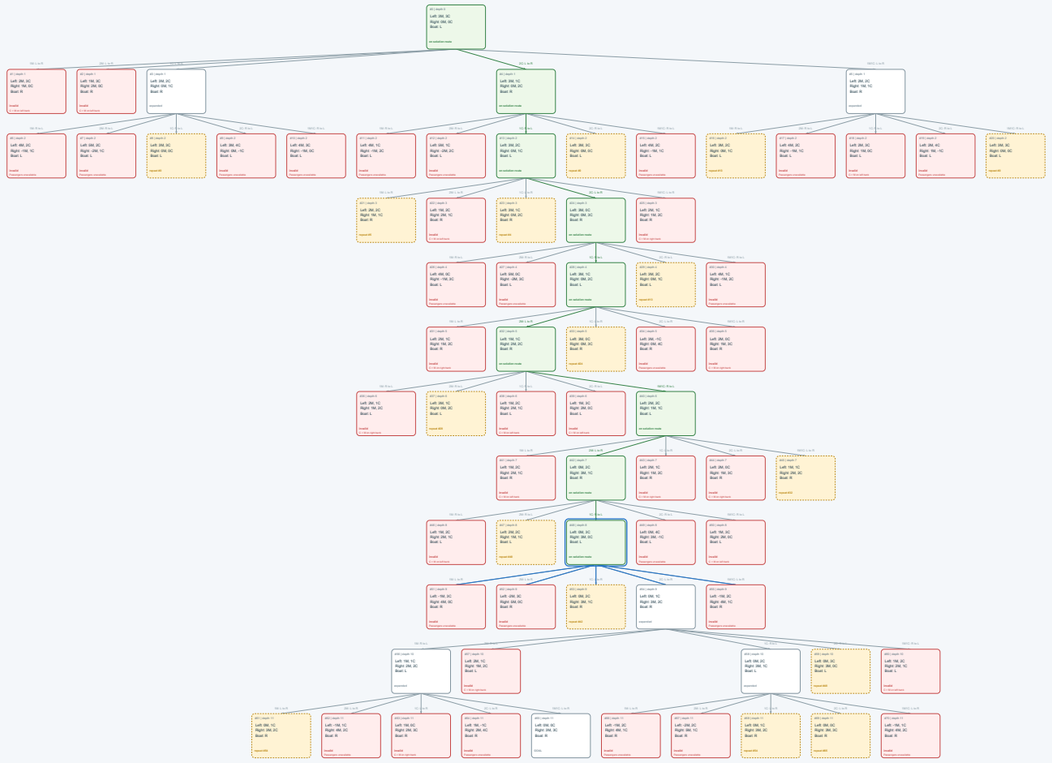
Move 2 missionaries from left to right. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 8 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



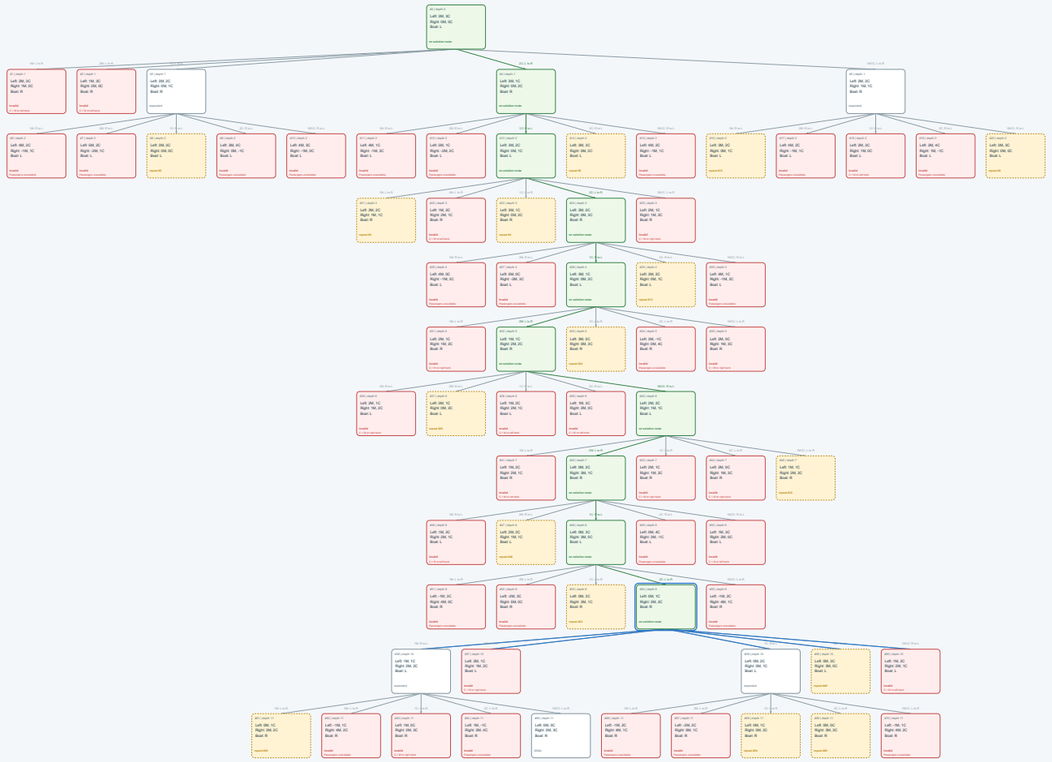
Move 1 cannibal from right to left. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 9 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#48 | depth 8

Left: 0M, 3C
Right: 3M, 0C
Boat: L

on solution route

2C: L to R

#54 | depth 9

Left: 0M, 1C
Right: 3M, 2C
Boat: R

on solution route

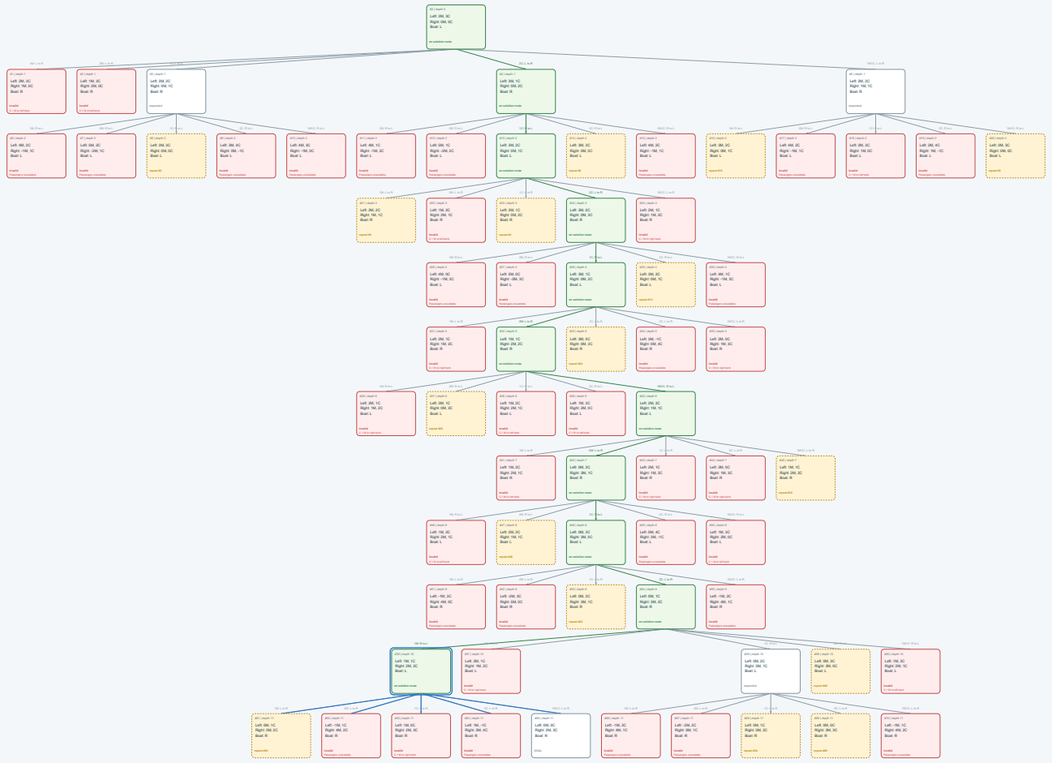
Move 2 cannibals from left to right. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 10 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above

#54 | depth 9

Left: 0M, 1C
Right: 3M, 2C
Boat: R

on solution route

1M: R to L

#56 | depth 10

Left: 1M, 1C
Right: 2M, 2C
Boat: L

on solution route

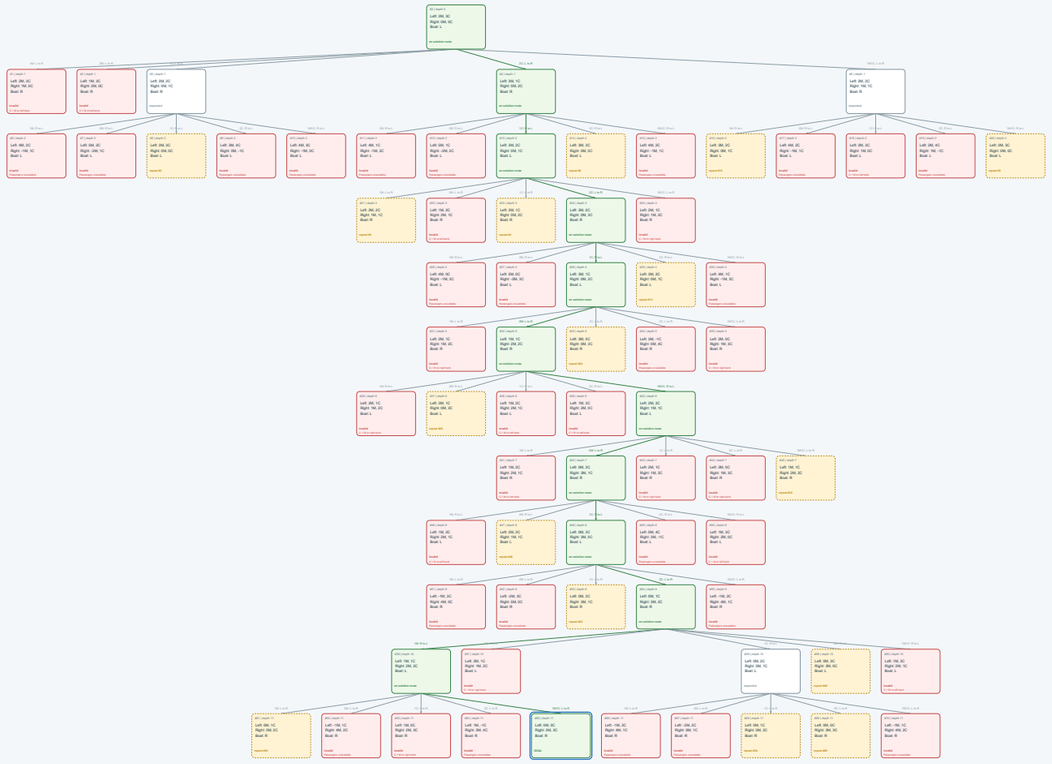
Move 1 missionary from right to left. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe.

Final solution - move 11 of 11

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Move 1 missionary and 1 cannibal from left to right. The boat changes bank. Its load is between 1 and 2 people, and both resulting banks are safe. The goal is now satisfied.

Final result and verification

The returned solution has been independently checked.

SOLVED - shortest depth 11

(3, 3, L) -> (3, 1, R) -> (3, 2, L) -> (3, 0, R) -> (3, 1, L) -> (1, 1, R) -> (2, 2, L) -> (0, 2, R) -> (0, 3, L) -> (0, 1, R) -> (1, 1, L) -> (0, 0, R)

VALIDATION

Every transition was independently checked. For 8-Puzzle this means one adjacent blank swap and preservation of the tiles. For the river problem this means an available boat load of 1-2, alternating boat side, safe banks and the exact final state.

WHY SHORTEST?

All moves cost one. BFS processes states in nondecreasing depth and checks the goal on dequeue. No shallower goal was processed before this one. The proof applies to the specified start, goal and action rules.

READING THE COUNTS

There are 15 unique discovered states, including the initial state. The 70 attempted transitions include 40 invalid moves and 16 repeats. Those leaf attempts are drawn for explanation but do not represent additional visited states.